Space restriction to the OBS virtual space

generated by `space_restriction.py`

In the CF-F12 construction of Yanai the virtual space splits into the OBS virtual space and the CABS+∞ remainder. We restrict the virtual space to OBS-vir: every external virtual index is an OBS-vir index, and the amplitude t_{ij}^{ab} vanishes when both of its indices are OBS-vir. In the many-body expansions of `many_body_equations.tex` (taken from the `expanded_blocks()` API of `expand_commutators.py`) this means:

- a virtual index contracted with another tensor is OK – the contraction sum still runs over the full virtual space;
- a term is zero iff both virtual indices of *one and the same* t_{ij}^{ab} amplitude sit on the operator $\hat{E}$ – the free indices of the operator are the external (OBS-vir) indices of the equation, so the term carries the vanishing amplitude. Two free virtual indices stemming from *different* amplitudes do not kill the term.

Each block is shown first as the original expansion with the vanishing terms underlined, then as the restricted expansion gathering the nonzero terms. Blocks still containing operators of rank higher than two are marked; their cumulant expansion is done later.

1 First-order commutators

$\left[\hat{h},\, \hat{T}_1\right]$ **(2 terms, 0 vanish)**

$$\left[\hat{h},\, \hat{T}_1\right] = -h_{p_1}^{i_1} t_{i_1}^{a_1} \hat{E}_{p_1}^{a_1} + h_{p_1}^{a_1} t_{i_1}^{a_1} \hat{E}_{i_1}^{p_1} \tag{1}$$

restricted:

$$\left[\hat{h},\, \hat{T}_1\right] = -h_{p_1}^{i_1} t_{i_1}^{a_1} \hat{E}_{p_1}^{a_1} + h_{p_1}^{a_1} t_{i_1}^{a_1} \hat{E}_{i_1}^{p_1} \tag{2}$$

$\left[\hat{h},\, \hat{T}_2\right]$ **(2 terms, 1 vanish)**

$$\left[\hat{h},\, \hat{T}_2\right] = \underline{-h_{p_1}^{i_2} t_{i_1 i_2}^{a_1 a_2} \hat{E}_{i_1 p_1}^{a_1 a_2}} \tag{3}$$

$$\underline{\qquad\qquad} + h_{p_1}^{a_2} t_{i_2 i_1}^{a_2 a_1} \hat{E}_{i_1 i_2}^{a_1 p_1} \tag{4}$$

restricted:

$$\left[\hat{h},\, \hat{T}_2\right] = h_{p_1}^{a_2} t_{i_2 i_1}^{a_2 a_1} \hat{E}_{i_1 i_2}^{a_1 p_1} \tag{5}$$

$\left[\hat{V}, \hat{T}_1\right]$ **(2 terms, 0 vanish)**

$$\left[\hat{V}, \hat{T}_1\right] = -g^{i_1 p_1}_{p_2 p_3} t^{a_1}_{i_1} \hat{E}^{a_1 p_1}_{p_2 p_3} \tag{6}$$
$$+ g^{a_1 p_3}_{p_1 p_2} t^{a_1}_{i_1} \hat{E}^{p_1 p_2}_{i_1 p_3} \tag{7}$$

restricted:

$$\left[\hat{V}, \hat{T}_1\right] = -g^{i_1 p_1}_{p_2 p_3} t^{a_1}_{i_1} \hat{E}^{a_1 p_1}_{p_2 p_3} \tag{8}$$
$$+ g^{a_1 p_3}_{p_1 p_2} t^{a_1}_{i_1} \hat{E}^{p_1 p_2}_{i_1 p_3} \tag{9}$$

$\left[\hat{V}, \hat{T}_2\right]$ **(4 terms, 2 vanish)**

$$\left[\hat{V}, \hat{T}_2\right] = \underline{-\tfrac{1}{2}\, g^{i_1 i_2}_{p_1 p_2} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{a_1 a_2}_{p_1 p_2}} \tag{10}$$
$$+ \tfrac{1}{2}\, g^{a_1 a_2}_{p_1 p_2} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{p_1 p_2}_{i_1 i_2} \tag{11}$$
$$\underline{-g^{i_2 p_1}_{p_2 p_3} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{a_1 a_2 p_1}_{i_1 p_2 p_3}} \tag{12}$$
$$+ g^{a_2 p_3}_{p_1 p_2} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{a_1 p_1 p_2}_{i_1 i_2 p_3} \tag{13}$$

restricted:

$$\left[\hat{V}, \hat{T}_2\right] = \tfrac{1}{2}\, g^{a_1 a_2}_{p_1 p_2} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{p_1 p_2}_{i_1 i_2} \tag{14}$$
$$+ g^{a_2 p_3}_{p_1 p_2} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{a_1 p_1 p_2}_{i_1 i_2 p_3} \quad \text{(three-body terms)} \tag{15}$$

2 Second order: Fock blocks

$\left[\left[\hat{F}, \hat{T}_1\right], \hat{T}_1\right]$ **(1 terms, 0 vanish)**

$$\left[\left[\hat{F}, \hat{T}_1\right], \hat{T}_1\right] = -2\, f^{a_2}_{i_2} t^{a_1}_{i_2} t^{a_2}_{i_1} \hat{E}^{a_1}_{i_1} \tag{16}$$

restricted:

$$\left[\left[\hat{F}, \hat{T}_1\right], \hat{T}_1\right] = -2\, f^{a_2}_{i_2} t^{a_1}_{i_2} t^{a_2}_{i_1} \hat{E}^{a_1}_{i_1} \tag{17}$$

$\left[\left[\hat{F}, \hat{T}_1\right], \hat{T}_2\right]$ **(2 terms, 1 vanish)**

$$\left[\left[\hat{F}, \hat{T}_1\right], \hat{T}_2\right] = \underline{-f^{a_3}_{i_3} t^{a_1 a_2}_{i_1 i_3} t^{a_3}_{i_2} \hat{E}^{a_1 a_2}_{i_1 i_2}} \tag{18}$$
$$- f^{a_3}_{i_3} t^{a_1}_{i_3} t^{a_3 a_2}_{i_1 i_2} \hat{E}^{a_1 a_2}_{i_1 i_2} \tag{19}$$

restricted:

$$\left[\left[\hat{F}, \hat{T}_1\right], \hat{T}_2\right] = -f^{a_3}_{i_3} t^{a_1}_{i_3} t^{a_3 a_2}_{i_1 i_2} \hat{E}^{a_1 a_2}_{i_1 i_2} \tag{20}$$

$\left[\left[\hat{F},\,\hat{T}_2\right],\,\hat{T}_1\right]$ **(2 terms, 1 vanish)**

$$\left[\left[\hat{F},\,\hat{T}_2\right],\,\hat{T}_1\right] = \underline{-f^{a_3}_{i_3} t^{a_1 a_2}_{i_1 i_3} t^{a_3}_{i_2} \hat{E}^{a_1 a_2}_{i_1 i_2}} \tag{21}$$

$$- f^{a_3}_{i_3} t^{a_1}_{i_3} t^{a_3 a_2}_{i_1 i_2} \hat{E}^{a_1 a_2}_{i_1 i_2} \tag{22}$$

restricted:

$$\left[\left[\hat{F},\,\hat{T}_2\right],\,\hat{T}_1\right] = -f^{a_3}_{i_3} t^{a_1}_{i_3} t^{a_3 a_2}_{i_1 i_2} \hat{E}^{a_1 a_2}_{i_1 i_2} \tag{23}$$

$\left[\left[\hat{F},\,\hat{T}_2\right],\,\hat{T}_2\right]$ **(1 terms, 1 vanish)**

$$\left[\left[\hat{F},\,\hat{T}_2\right],\,\hat{T}_2\right] = \underline{-2\, f^{a_4}_{i_4} t^{a_1 a_2}_{i_1 i_4} t^{a_4 a_3}_{i_2 i_3} \hat{E}^{a_1 a_2 a_3}_{i_1 i_2 i_3}} \tag{24}$$

restricted:

$$\left[\left[\hat{F},\,\hat{T}_2\right],\,\hat{T}_2\right] = 0 \tag{25}$$

$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_1\right]$ **(4 terms, 0 vanish)**

$$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_1\right] = f^{i_1}_{i_2} t^{a_1}_{i_1} t^{a_2}_{i_2} \hat{E}^{a_1}_{a_2} \tag{26}$$

$$- f^{a_2}_{p_1} t^{a_1}_{i_1} t^{a_2}_{i_1} \hat{E}^{a_1}_{p_1} + f^{a_1}_{a_2} t^{a_1}_{i_1} t^{a_2}_{i_2} \hat{E}^{i_1}_{i_2} \tag{27}$$

$$- f^{i_2}_{p_1} t^{a_1}_{i_1} t^{a_1}_{i_2} \hat{E}^{p_1}_{i_1} \tag{28}$$

restricted:

$$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_1\right] = f^{i_1}_{i_2} t^{a_1}_{i_1} t^{a_2}_{i_2} \hat{E}^{a_1}_{a_2} \tag{29}$$

$$- f^{a_2}_{p_1} t^{a_1}_{i_1} t^{a_2}_{i_1} \hat{E}^{a_1}_{p_1} + f^{a_1}_{a_2} t^{a_1}_{i_1} t^{a_2}_{i_2} \hat{E}^{i_1}_{i_2} \tag{30}$$

$$- f^{i_2}_{p_1} t^{a_1}_{i_1} t^{a_1}_{i_2} \hat{E}^{p_1}_{i_1} \tag{31}$$

$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_2\right]$ **(4 terms, 2 vanish)**

$$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_2\right] = \underline{f^{i_2}_{i_3} t^{a_1 a_2}_{i_2 i_1} t^{a_3}_{i_3} \hat{E}^{a_1 a_2}_{a_3 i_1}} \tag{32}$$

$$\underline{-f^{a_3}_{p_1} t^{a_1 a_2}_{i_1 i_2} t^{a_3}_{i_2} \hat{E}^{a_1 a_2}_{i_1 p_1}} \tag{33}$$

$$+ f^{a_2}_{a_3} t^{a_2 a_1}_{i_3 i_2} t^{a_3}_{i_1} \hat{E}^{a_1 i_1}_{i_2 i_3} \tag{34}$$

$$- f^{i_3}_{p_1} t^{a_2 a_1}_{i_2 i_1} t^{a_2}_{i_3} \hat{E}^{a_1 p_1}_{i_1 i_2} \tag{35}$$

restricted:

$$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_2\right] = f^{a_2}_{a_3} t^{a_2 a_1}_{i_3 i_2} t^{a_3}_{i_1} \hat{E}^{a_1 i_1}_{i_2 i_3} \tag{36}$$

$$- f^{i_3}_{p_1} t^{a_2 a_1}_{i_2 i_1} t^{a_2}_{i_3} \hat{E}^{a_1 p_1}_{i_1 i_2} \tag{37}$$

$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_1\right]$ **(8 terms, 2 vanish)**

$$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_1\right] = \underline{f^{i_2}_{i_3} t^{a_1}_{i_2} t^{a_2 a_3}_{i_3 i_1} \hat{E}^{a_1 i_1}_{a_2 a_3}} \tag{38}$$

$$- f^{a_3}_{p_1} t^{a_1}_{i_2} t^{a_3 a_2}_{i_1 i_2} \hat{E}^{a_1 i_1}_{a_2 p_1} \tag{39}$$

$$- f^{a_3}_{p_1} t^{a_1}_{i_2} t^{a_3 a_2}_{i_2 i_1} \hat{E}^{a_1 i_1}_{p_1 a_2} \tag{40}$$

$$\underline{+ f^{i_1}_{p_1} t^{a_1}_{i_2} t^{a_2 a_3}_{i_2 i_1} \hat{E}^{a_1 p_1}_{a_2 a_3}} \tag{41}$$

$$+ f^{a_2}_{a_3} t^{a_2 a_1}_{i_2 i_1} t^{a_3}_{i_3} \hat{E}^{i_1 i_2}_{a_1 i_3} \tag{42}$$

$$+ f^{a_1}_{p_1} t^{a_1 a_2}_{i_2 i_1} t^{a_2}_{i_3} \hat{E}^{i_1 i_2}_{i_3 p_1} \tag{43}$$

$$- f^{i_3}_{p_1} t^{a_2 a_1}_{i_3 i_1} t^{a_2}_{i_2} \hat{E}^{i_1 p_1}_{a_1 i_2} \tag{44}$$

$$- f^{i_3}_{p_1} t^{a_2 a_1}_{i_1 i_3} t^{a_2}_{i_2} \hat{E}^{i_1 p_1}_{i_2 a_1} \tag{45}$$

restricted:

$$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_1\right] = - f^{a_3}_{p_1} t^{a_1}_{i_2} t^{a_3 a_2}_{i_1 i_2} \hat{E}^{a_1 i_1}_{a_2 p_1} \tag{46}$$

$$- f^{a_3}_{p_1} t^{a_1}_{i_2} t^{a_3 a_2}_{i_2 i_1} \hat{E}^{a_1 i_1}_{p_1 a_2} \tag{47}$$

$$+ f^{a_2}_{a_3} t^{a_2 a_1}_{i_2 i_1} t^{a_3}_{i_3} \hat{E}^{i_1 i_2}_{a_1 i_3} \tag{48}$$

$$+ f^{a_1}_{p_1} t^{a_1 a_2}_{i_2 i_1} t^{a_2}_{i_3} \hat{E}^{i_1 i_2}_{i_3 p_1} \tag{49}$$

$$- f^{i_3}_{p_1} t^{a_2 a_1}_{i_3 i_1} t^{a_2}_{i_2} \hat{E}^{i_1 p_1}_{a_1 i_2} \tag{50}$$

$$- f^{i_3}_{p_1} t^{a_2 a_1}_{i_1 i_3} t^{a_2}_{i_2} \hat{E}^{i_1 p_1}_{i_2 a_1} \tag{51}$$

$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_2\right]$ **(12 terms, 6 vanish)**

$$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_2\right] = \underline{f^{i_1}_{i_2} t^{a_1 a_2}_{i_1 i_3} t^{a_3 a_4}_{i_2 i_3} \hat{E}^{a_1 a_2}_{a_3 a_4}} \tag{52}$$

$$\underline{- f^{a_4}_{p_1} t^{a_1 a_2}_{i_1 i_2} t^{a_4 a_3}_{i_2 i_1} \hat{E}^{a_1 a_2}_{a_3 p_1}} \tag{53}$$

$$+ f^{a_1}_{a_2} t^{a_1 a_3}_{i_1 i_2} t^{a_2 a_3}_{i_3 i_4} \hat{E}^{i_1 i_2}_{i_3 i_4} \tag{54}$$

$$- f^{i_4}_{p_1} t^{a_1 a_2}_{i_1 i_4} t^{a_1 a_2}_{i_2 i_3} \hat{E}^{i_1 p_1}_{i_2 i_3} \tag{55}$$

$$\underline{+ f^{i_3}_{i_4} t^{a_1 a_2}_{i_3 i_2} t^{a_3 a_4}_{i_4 i_1} \hat{E}^{a_1 a_2 i_1}_{a_3 i_2 a_4}} \tag{56}$$

$$\underline{- f^{a_4}_{p_1} t^{a_1 a_2}_{i_3 i_2} t^{a_4 a_3}_{i_1 i_3} \hat{E}^{a_1 a_2 i_1}_{a_3 i_2 p_1}} \tag{57}$$

$$\underline{- f^{a_4}_{p_1} t^{a_1 a_2}_{i_2 i_3} t^{a_4 a_3}_{i_3 i_1} \hat{E}^{a_1 a_2 i_1}_{i_2 p_1 a_3}} \tag{58}$$

$$\underline{+ f^{i_2}_{p_1} t^{a_1 a_2}_{i_3 i_1} t^{a_3 a_4}_{i_3 i_2} \hat{E}^{a_1 a_2 p_1}_{a_3 i_1 a_4}} \tag{59}$$

$$+ f^{a_3}_{a_4} t^{a_3 a_1}_{i_4 i_3} t^{a_4 a_2}_{i_2 i_1} \hat{E}^{a_1 i_1 i_2}_{i_3 a_2 i_4} \tag{60}$$

$$+ f^{a_2}_{p_1} t^{a_2 a_3}_{i_2 i_1} t^{a_3 a_1}_{i_4 i_3} \hat{E}^{a_1 i_1 i_2}_{i_3 i_4 p_1} \tag{61}$$

$$- f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_2} t^{a_3 a_2}_{i_4 i_1} \hat{E}^{a_1 i_1 p_1}_{i_2 a_2 i_3} \tag{62}$$

$$- f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_2} t^{a_3 a_2}_{i_1 i_4} \hat{E}^{a_1 i_1 p_1}_{i_2 i_3 a_2} \tag{63}$$

restricted:

$$\left[\left[\hat{F},\, \hat{T}_2\right]^\dagger,\, \hat{T}_2\right] = f^{a_1}_{a_2} t^{a_1 a_3}_{i_1 i_2} t^{a_2 a_3}_{i_3 i_4} \hat{E}^{i_1 i_2}_{i_3 i_4} \tag{64}$$

$$- f^{i_4}_{p_1} t^{a_1 a_2}_{i_1 i_4} t^{a_1 a_2}_{i_2 i_3} \hat{E}^{i_1 p_1}_{i_2 i_3} \tag{65}$$

$$+ f^{a_3}_{a_4} t^{a_3 a_1}_{i_4 i_3} t^{a_4 a_2}_{i_2 i_1} \hat{E}^{a_1 i_1 i_2}_{i_3 a_2 i_4} \tag{66}$$

$$+ f^{a_2}_{p_1} t^{a_2 a_3}_{i_2 i_1} t^{a_3 a_1}_{i_4 i_3} \hat{E}^{a_1 i_1 i_2}_{i_3 i_4 p_1} \tag{67}$$

$$- f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_2} t^{a_3 a_2}_{i_4 i_1} \hat{E}^{a_1 i_1 p_1}_{i_2 a_2 i_3} \tag{68}$$

$$- f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_2} t^{a_3 a_2}_{i_1 i_4} \hat{E}^{a_1 i_1 p_1}_{i_2 i_3 a_2} \quad \text{(three-body terms)} \tag{69}$$

3 The full expression in the OBS-vir restriction

The commutator stage assembles the blocks into the transcorrelated Hamiltonian through $O(\lambda^2)$ as the blocks plus their Hermitian conjugates (the anti-Hermitian generator $\hat{A} = \hat{T} - \hat{T}^\dagger$), the Fock double-commutator blocks carrying the factor $1/2$ (see the Assembly section of `many_body_equations.tex`). Assembling the *restricted* blocks the same way and merging over the exact identities (conjugate pairs with equal canonical form fold into one term, self-conjugate terms keeping coefficient 1 through the $1/2$) gives the transcorrelated Hamiltonian in the OBS-vir restriction:

$$\bar{H} - \hat{H} = f^{i_1}_{i_2} t^{a_1}_{i_1} t^{a_2}_{i_2} \hat{E}^{a_1}_{a_2} \tag{70}$$

$$- f^{a_2}_{i_2} t^{a_1}_{i_2} t^{a_2}_{i_1} \hat{E}^{a_1}_{i_1} \tag{71}$$

$$- \tfrac{1}{2} f^{a_2}_{p_1} t^{a_1}_{i_1} t^{a_2}_{i_1} \hat{E}^{a_1}_{p_1} - h^{i_1}_{p_1} t^{a_1}_{i_1} \hat{E}^{a_1}_{p_1} \tag{72}$$

$$- f^{a_2}_{i_2} t^{a_1}_{i_2} t^{a_2}_{i_1} \hat{E}^{i_1}_{a_1} + f^{a_1}_{a_2} t^{a_1}_{i_1} t^{a_2}_{i_2} \hat{E}^{i_1}_{i_2} \tag{73}$$

$$- \tfrac{1}{2} f^{i_2}_{p_1} t^{a_1}_{i_1} t^{a_1}_{i_2} \hat{E}^{i_1}_{p_1} + h^{a_1}_{p_1} t^{a_1}_{i_1} \hat{E}^{i_1}_{p_1} \tag{74}$$

$$- \tfrac{1}{2} f^{a_2}_{p_1} t^{a_1}_{i_1} t^{a_2}_{i_1} \hat{E}^{p_1}_{a_1} - h^{i_1}_{p_1} t^{a_1}_{i_1} \hat{E}^{p_1}_{a_1} \tag{75}$$

$$- \tfrac{1}{2} f^{i_2}_{p_1} t^{a_1}_{i_1} t^{a_1}_{i_2} \hat{E}^{p_1}_{i_1} + h^{a_1}_{p_1} t^{a_1}_{i_1} \hat{E}^{p_1}_{i_1} \tag{76}$$

$$- f^{a_3}_{i_3} t^{a_1}_{i_3} t^{a_3 a_2}_{i_1 i_2} \hat{E}^{a_1 a_2}_{i_1 i_2} \tag{77}$$

$$- \tfrac{1}{2} f^{a_3}_{p_1} t^{a_1}_{i_2} t^{a_3 a_2}_{i_1 i_2} \hat{E}^{a_1 i_1}_{a_2 p_1} \tag{78}$$

$$+ f^{a_2}_{a_3} t^{a_2 a_1}_{i_3 i_2} t^{a_3}_{i_1} \hat{E}^{a_1 i_1}_{i_2 i_3} \tag{79}$$

$$- \tfrac{1}{2} f^{i_3}_{p_1} t^{a_2 a_1}_{i_3 i_2} t^{a_2}_{i_1} \hat{E}^{a_1 i_1}_{i_2 p_1} \tag{80}$$

$$- \tfrac{1}{2} f^{a_3}_{p_1} t^{a_1}_{i_2} t^{a_3 a_2}_{i_2 i_1} \hat{E}^{a_1 i_1}_{p_1 a_2} \tag{81}$$

$$- \tfrac{1}{2} f^{i_3}_{p_1} t^{a_2 a_1}_{i_2 i_3} t^{a_2}_{i_1} \hat{E}^{a_1 i_1}_{p_1 i_2} \tag{82}$$

$$- \tfrac{1}{2} f^{a_3}_{p_1} t^{a_2}_{i_2} t^{a_3 a_1}_{i_1 i_2} \hat{E}^{a_1 p_1}_{a_2 i_1} \tag{83}$$

$$- \tfrac{1}{2} f^{a_3}_{p_1} t^{a_2}_{i_2} t^{a_3 a_1}_{i_2 i_1} \hat{E}^{a_1 p_1}_{i_1 a_2} \tag{84}$$

$$- \tfrac{1}{2} f^{i_3}_{p_1} t^{a_2 a_1}_{i_2 i_1} t^{a_2}_{i_3} \hat{E}^{a_1 p_1}_{i_1 i_2} \tag{85}$$

$$+ h^{a_2}_{p_1} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{a_1 p_1}_{i_1 i_2} - g^{i_1 p_1}_{p_2 p_3} t^{a_1}_{i_1} \hat{E}^{a_1 p_1}_{p_2 p_3} \tag{86}$$

$$- f^{a_3}_{i_3} t^{a_1}_{i_3} t^{a_3 a_2}_{i_1 i_2} \hat{E}^{i_1 i_2}_{a_1 a_2} \tag{87}$$

$$+ f^{a_2}_{a_3} t^{a_2 a_1}_{i_2 i_1} t^{a_3}_{i_3} \hat{E}^{i_1 i_2}_{a_1 i_3} \tag{88}$$

$$- \tfrac{1}{2}\, f^{i_3}_{p_1} t^{a_2 a_1}_{i_2 i_1} t^{a_2}_{i_3} \hat{E}^{i_1 i_2}_{a_1 p_1} \tag{89}$$
$$+ h^{a_2}_{p_1} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{i_1 i_2}_{a_1 p_1} \tag{90}$$
$$+ f^{a_1}_{a_2} t^{a_1 a_3}_{i_1 i_2} t^{a_2 a_3}_{i_3 i_4} \hat{E}^{i_1 i_2}_{i_3 i_4} \tag{91}$$
$$+ \tfrac{1}{2}\, f^{a_1}_{p_1} t^{a_1 a_2}_{i_2 i_1} t^{a_2}_{i_3} \hat{E}^{i_1 i_2}_{i_3 p_1} \tag{92}$$
$$- \tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_1 a_2}_{i_1 i_2} t^{a_1 a_2}_{i_3 i_4} \hat{E}^{i_1 i_2}_{i_3 p_1} \tag{93}$$
$$+ \tfrac{1}{2}\, g^{a_1 a_2}_{p_1 p_2} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{i_1 i_2}_{p_1 p_2} \tag{94}$$
$$- \tfrac{1}{2}\, f^{i_3}_{p_1} t^{a_2 a_1}_{i_3 i_1} t^{a_2}_{i_2} \hat{E}^{i_1 p_1}_{a_1 i_2} \tag{95}$$
$$- \tfrac{1}{2}\, f^{i_3}_{p_1} t^{a_2 a_1}_{i_1 i_3} t^{a_2}_{i_2} \hat{E}^{i_1 p_1}_{i_2 a_1} \tag{96}$$
$$+ \tfrac{1}{2}\, f^{a_1}_{p_1} t^{a_1 a_2}_{i_3 i_2} t^{a_2}_{i_1} \hat{E}^{i_1 p_1}_{i_2 i_3} \tag{97}$$
$$- \tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_1 a_2}_{i_1 i_4} t^{a_1 a_2}_{i_2 i_3} \hat{E}^{i_1 p_1}_{i_2 i_3} \tag{98}$$
$$+ g^{a_1 p_1}_{p_2 p_3} t^{a_1}_{i_1} \hat{E}^{i_1 p_1}_{p_2 p_3} - g^{i_1 p_3}_{p_1 p_2} t^{a_1}_{i_1} \hat{E}^{p_1 p_2}_{a_1 p_3} \tag{99}$$
$$+ \tfrac{1}{2}\, g^{a_1 a_2}_{p_1 p_2} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{p_1 p_2}_{i_1 i_2} \tag{100}$$
$$+ g^{a_1 p_3}_{p_1 p_2} t^{a_1}_{i_1} \hat{E}^{p_1 p_2}_{i_1 p_3} \tag{101}$$
$$+ f^{a_3}_{a_4} t^{a_3 a_1}_{i_4 i_3} t^{a_4 a_2}_{i_2 i_1} \hat{E}^{a_1 i_1 i_2}_{i_3 a_2 i_4} \tag{102}$$
$$- \tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_3 a_1}_{i_4 i_3} t^{a_3 a_2}_{i_2 i_1} \hat{E}^{a_1 i_1 i_2}_{i_3 a_2 p_1} \tag{103}$$
$$+ \tfrac{1}{2}\, f^{a_2}_{p_1} t^{a_2 a_3}_{i_2 i_1} t^{a_3 a_1}_{i_4 i_3} \hat{E}^{a_1 i_1 i_2}_{i_3 i_4 p_1} \tag{104}$$
$$- \tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_4} t^{a_3 a_2}_{i_2 i_1} \hat{E}^{a_1 i_1 i_2}_{p_1 a_2 i_3} \tag{105}$$
$$- \tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_2} t^{a_3 a_2}_{i_4 i_1} \hat{E}^{a_1 i_1 p_1}_{i_2 a_2 i_3} \tag{106}$$
$$- \tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_2} t^{a_3 a_2}_{i_1 i_4} \hat{E}^{a_1 i_1 p_1}_{i_2 i_3 a_2} \tag{107}$$
$$+ g^{a_2 p_3}_{p_1 p_2} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{a_1 p_1 p_2}_{i_1 i_2 p_3} \tag{108}$$
$$+ \tfrac{1}{2}\, f^{a_2}_{p_1} t^{a_2 a_3}_{i_4 i_3} t^{a_3 a_1}_{i_2 i_1} \hat{E}^{i_1 i_2 p_1}_{a_1 i_3 i_4} \tag{109}$$
$$+ g^{a_2 p_1}_{p_2 p_3} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{i_1 i_2 p_1}_{a_1 p_2 p_3} \tag{110}$$

Keeping only the pure-T_2 blocks (the F12 geminal operator $\hat{A}^{F12} = \hat{T}_2 - \hat{T}_2^\dagger$; the geminal amplitudes carry the pair-exchange symmetry $G^{\alpha\beta}_{ij} = G^{ij}_{\alpha\beta}$ and vanish with both indices OBS-vir) this is eq. (S1) of the Supplemental Material of Masteran et al., J. Chem. Phys. 158, 057101 (2023), the complete transcorrelated Hamiltonian obtained without the cumulant decomposition – verified term by term and numerically in `test/check_reference_s1.py`:

$$\bar{H}^{F12} - \hat{H} = h^{a_2}_{p_1} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{a_1 p_1}_{i_1 i_2} \tag{111}$$
$$+ h^{a_2}_{p_1} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{i_1 i_2}_{a_1 p_1} \tag{112}$$
$$+ f^{a_1}_{a_2} t^{a_1 a_3}_{i_1 i_2} t^{a_2 a_3}_{i_3 i_4} \hat{E}^{i_1 i_2}_{i_3 i_4} \tag{113}$$
$$- \tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_1 a_2}_{i_1 i_2} t^{a_1 a_2}_{i_3 i_4} \hat{E}^{i_1 i_2}_{i_3 p_1} \tag{114}$$
$$+ \tfrac{1}{2}\, g^{a_1 a_2}_{p_1 p_2} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{i_1 i_2}_{p_1 p_2} \tag{115}$$
$$- \tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_1 a_2}_{i_1 i_4} t^{a_1 a_2}_{i_2 i_3} \hat{E}^{i_1 p_1}_{i_2 i_3} \tag{116}$$
$$+ \tfrac{1}{2}\, g^{a_1 a_2}_{p_1 p_2} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{p_1 p_2}_{i_1 i_2} \tag{117}$$
$$+ f^{a_3}_{a_4} t^{a_3 a_1}_{i_4 i_3} t^{a_4 a_2}_{i_2 i_1} \hat{E}^{a_1 i_1 i_2}_{i_3 a_2 i_4} \tag{118}$$

$$- \tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_3 a_1}_{i_4 i_3} t^{a_3 a_2}_{i_2 i_1} \hat{E}^{a_1 i_1 i_2}_{i_3 a_2 p_1} \tag{119}$$

$$+ \tfrac{1}{2}\, f^{a_2}_{p_1} t^{a_2 a_3}_{i_2 i_1} t^{a_3 a_1}_{i_4 i_3} \hat{E}^{a_1 i_1 i_2}_{i_3 i_4 p_1} \tag{120}$$

$$- \tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_4} t^{a_3 a_2}_{i_2 i_1} \hat{E}^{a_1 i_1 i_2}_{p_1 a_2 i_3} \tag{121}$$

$$- \tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_2} t^{a_3 a_2}_{i_4 i_1} \hat{E}^{a_1 i_1 p_1}_{i_2 a_2 i_3} \tag{122}$$

$$- \tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_2} t^{a_3 a_2}_{i_1 i_4} \hat{E}^{a_1 i_1 p_1}_{i_2 i_3 a_2} \tag{123}$$

$$+ g^{a_2 p_3}_{p_1 p_2} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{a_1 p_1 p_2}_{i_1 i_2 p_3} \tag{124}$$

$$+ \tfrac{1}{2}\, f^{a_2}_{p_1} t^{a_2 a_3}_{i_4 i_3} t^{a_3 a_1}_{i_2 i_1} \hat{E}^{i_1 i_2 p_1}_{a_1 i_3 i_4} \tag{125}$$

$$+ g^{a_2 p_1}_{p_2 p_3} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{i_1 i_2 p_1}_{a_1 p_2 p_3} \tag{126}$$